In a time t seconds, the coil is rotated through an angle θ (= ωt) in anti–clockwise direction. In this position, the flux linked $NBA\cos\omega t$ is due to the component of $\vec{B}$ normal to the plane of the coil. The component ($B\sin\omega t$) parallel to the plane has no role in electromagnetic induction. Therefore, the flux linkage with the coil at this deflected position is

$$N\Phi_B = NBA\cos\theta = NBA\cos\omega t$$

According to Faraday's law, the emf induced at that instant is

$$\begin{aligned}\varepsilon &= -\frac{d}{dt}(N\Phi_B) = -\frac{d}{dt}(NBA\cos\omega t)\\ &= -NBA(-\sin\omega t)\omega\\ &= NBA\,\omega\sin\omega t\end{aligned}$$

When the coil is rotated through 90° from initial position, $\sin\omega t$ = 1. Then the maximum value of induced emf is

$$\varepsilon_m = NBA\omega$$

Therefore, the value of induced emf at any instant is then given by

$$\varepsilon = \varepsilon_m \sin\omega t \qquad (4.22)$$

It is seen that the induced emf varies as sine function of the time angle ωt. The graph between induced emf and time angle for one rotation of the coil will be a sine curve (Figure 4.25) and the emf varying in this manner is called **sinusoidal emf** or **alternating emf**.

If this alternating voltage is given to a closed circuit, a sinusoidally varying current flows in it. This current is called **alternating current** and is given by

$$i = I_m \sin\omega t \qquad (4.23)$$

where I_m is the maximum value of induced current.

EXAMPLE 4.15

A rectangular coil of area 70 cm^2 having 600 turns rotates about an axis perpendicular to a magnetic field of 0.4 Wb m^{-2}. If the coil completes 500 revolutions in a minute, calculate the instantaneous emf when the plane of the coil is (i) perpendicular to the field (ii) parallel to the field and (iii) inclined at 60° with the field.

Solution

$A = 70 \times 10^{-4}$ m^2; $N = 600$ turns

$B = 0.4$ Wbm^{-2}; $f = 500$ rpm

The instantaneous emf is

$$\varepsilon = \varepsilon_m \sin\omega t$$

$$\text{since } \varepsilon_m = N\Phi_m\omega = N(BA)(2\pi f)$$

$$\varepsilon = NBA \times 2\pi f \times \sin\omega t$$

(i) When $\omega t = 0^\circ$,

$$\varepsilon = \varepsilon_m \sin 0 = 0$$

(ii) When $\omega t = 90^\circ$,

$$\begin{aligned}\varepsilon = \varepsilon_m \sin 90^\circ &= NBA \times 2\pi f \times 1\\ &= 600 \times 0.4 \times 70 \times 10^{-4} \times 2 \times \frac{22}{7} \times \left(\frac{500}{60}\right)\\ &= 88\text{ V}\end{aligned}$$

(iii) When ωt = 90° – 60° = 30°,

$$\varepsilon = \varepsilon_m \sin 30^\circ = 88 \times \frac{1}{2} = 44\text{ V}$$

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